

## VAISHNAVI ASURI

Denver, CO (Open to Relocating)

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### SUMMARY

Versatile **Data Analyst** professional with **3+ years** of experience, proven success driving operational efficiency and automation across **healthcare** and **enterprise project management**. Expert in **Power BI, SQL, Python, AWS**, and Microsoft 365, with a strong track record of building scalable **ETL pipelines** and **AI-powered dashboards**. Reduced manual report lookup and cross-system navigation time by **50–60%** and cut repetitive IT support tickets by **~30%** through Copilot-style chatbot and RAG solutions. Streamlined vendor management and budget tracking for **300+ projects** while supporting **HIPAA/PII** compliance. **Actively seeking** to contribute to innovative projects within a dynamic and forward-thinking organization

### TECHNICAL SKILLS

- **Databases:** Microsoft SQL Server, MySQL, AWS Athena, AWS S3, Apache Hudi, ElasticSearch
- **Programming Languages:** Python (NumPy, Pandas, Scikit-learn, Matplotlib, PySpark), R, Power Query, DAX, SQL
- **Tools & Platforms:** Power BI, Tableau, Power Automate, Microsoft Fabric, Microsoft Planner, Project Web App (PWA), Microsoft 365, SharePoint, Smartsheet, Jira, Confluence, AWS (SageMaker, EMR, CloudWatch), OpenAI GPT-4 (via Azure, OpenAI), LangChain, FAISS Vector DB
- **Data Visualization:** Power BI, Tableau, Excel, PowerPoint
- **Machine Learning & Statistics:** Supervised ML (Logistic Regression, Decision Trees, Gradient Boost, SVM), Unsupervised ML (K-Means, Naïve Bayes, Association Rule Mining), NLP (SpaCy), T-SNE, LSTM, ARIMA, SARIMAX, Experimental Design, Bayesian Inference, Hierarchical Modeling, Longitudinal Data Analysis
- **ETL & Data Engineering:** Airflow, Data Warehousing, Metadata-Driven Pipelines, Data Modeling
- **Other:** Natural Language Processing, Quality Measurement, Healthcare Analytics, Accessibility Data, Risk Scoring, A/B Testing, Python DAX parser

### PROFESSIONAL CERTIFICATIONS AND ACHIEVEMENTS

- [Neo4j Fundamentals](#)
- [Identifying Database Workloads](#)
- [Data Warehousing](#)
- [Shell Programming](#)
- [Modeling Data Relationships](#)
- [Intermediate Python](#)
- [Power BI for Data Analysts](#)

### PROFESSIONAL EXPERIENCE

#### STATE OF COLORADO HEALTH CARE POLICY AND FINANCING

Denver, CO

##### Data Analyst

Nov 2024 - Present

*Support analytics and reporting for 300+ active projects tracked via Microsoft's Project Web App (PWA) by building dashboards and automations that streamline vendor management, budget tracking, and project oversight across teams and tools at the Enterprise Project Management Office while practicing HIPAA and PII compliance*

- Built a PoC for a Copilot-style chatbot integrated with **OpenAI's GPT-4**, **Microsoft Teams**, and PWA project metadata to answer natural language queries related to the status of the project/vendor to demonstrate potential to reduce manual report lookup and cross-system navigation time by 50–60% for project managers and analysts
- Optimized resource usage and cost efficiency by configuring **Azure** Monitor and Autoscale settings for dynamic workload management.
- Developed a **PoC** for a knowledge retrieval chatbot using **RAG (Retrieval-Augmented Generation)** architecture to address frequently asked questions related to **SharePoint**, **PWA**, **Power BI**, and **Smartsheet** tasks. The bot indexes

internal help documentation, training manuals, and IT ticket logs to provide instant answers. Currently expanding the knowledge base to cover broader user scenarios, with initial tests indicating potential to reduce repetitive IT support tickets by ~30%

- Built a metadata-aware **Power BI** dashboard documentation generator using a Copilot-style agent that auto-generates descriptions of measures, dimensions, filters, and their logic using **DAX queries** + comments, supporting onboarding and reducing documentation gaps
- Developed and **automated** enterprise-wide **reporting tools** and dashboards (**Power BI, Smartsheet**), streamlining vendor management, project tracking, and budget monitoring across multiple projects and vendors (**Microsoft 365, Power Platform, PWA**)
- Developed a **Vendor Management Scorecard & Dashboard** from scratch in **Power BI**, tracking 6 vendors and 20+ projects by extracting data from **Microsoft PWA**, enabling granular performance monitoring, and enforcing stricter vendor accountability metrics
- Automated invoice and budget tracking for 15+ vendors by designing a **Google Form-based data collection system**, eliminating manual email processing, and creating a dashboard to analyze payment timelines, budget utilization, and invoice trends for leadership reporting
- Led the automation of a Strategic **Project Tracker** in **Smartsheet**, consolidating data from **PWA** and team lead surveys for 180+ projects, reducing manual entry and improving monthly reporting efficiency
- Managed and optimized multiple **EPMO dashboards**, including a resource allocation tracker (staffing, hours, assignments) and a project performance dashboard tracking 200+ projects across risks, issues, and timelines for executive review
- Owned SharePoint administration and PWA support, maintaining internal/external project sites, troubleshooting access/issues, and training staff on platform use, while handling ad-hoc requests for 70+ FY closing projects and ongoing initiatives

## COHERE HEALTH

### Data Analyst /Data Engineer

Boston, MA

Nov 2023 - May 2024

*Worked on processing and preparing healthcare claims data from hospitals and providers to support machine learning models, signal-based LLMs, and automate claims workflows, with a strong focus on data quality, fraud detection, and HIPAA/PII compliance*

- Built scalable, metadata-driven **ETL pipelines** on **AWS** to **ingest, transform**, and centralize **healthcare** and **product data** from diverse sources, enabling analytics, quality control, and cross-functional reporting
- Built a PoC using an autoencoder-based anomaly detection model within the claims preprocessing pipeline to flag outliers or inconsistencies in otherwise clean claims data, allowing early verification before reaching downstream **ML/NLP models** and improving overall data quality assurance.
- Integrated an **XGBoost** model into a metadata-driven data validation **pipeline**, designed to assist the actuarial team by predicting expected value distributions (e.g., claim amounts, diagnosis-procedure pairings) and flagging entries that deviate from historical norms, enhancing the accuracy of downstream analysis and financial modeling
- Implemented scalable data ingestion pipeline using **PyGSheets** on **Sagemaker** and **Airflow** by reading from data sources including **ElasticSearch, Apache Hudi, S3, Google Sheets**, etc, to provide reports for research teams
- Designed and built abstract QC processes for healthcare data from multiple payors
- Ingested 600,000 records of healthcare benchmark data using metadata-driven transformation for downstream teams
- Reduced the production environment issues by performing regular on-call duty using **CloudWatch, Airflow**, and **Elastic MapReduce**, and provided constructive solutions to the team
- Identified product data bugs by **exploring data** and **collaborated** with the **Product** and **Engineer teams** to establish solutions by collaborating with the BI team
- Developed and managed **ETL pipelines** using **Airflow** and the **AWS** cloud platform to maintain data integrity and validity from diverse payers for analysis and predictive purposes
- Established a centralized data supermarket for downstream users to ensure a single source of truth across departments, thereby enhancing data integrity and accessibility
- Built scalable production jobs to ingest data coming from various customer feedback forms for the **Reporting** and **Quality Assessment teams**

## VIRTUSA

### Data Science Intern

Hyderabad, India

Jan 2022 - June 2022

*Worked as a Data Science Intern at Virtusa, gaining hands-on experience in data analysis, machine learning, and reporting tools. Focused on building foundational skills in Python, SQL, and data visualization while supporting real-world business analytics tasks*

- **Queried structured data** using **SQL** to generate performance and operations reports; supported business teams with insights extracted from **MySQL databases**.
- Created interactive dashboards and visualizations using **Tableau** and **Power BI** to present **KPIs** and trends for internal stakeholders
- Cleaned and preprocessed raw datasets using **Python (Pandas, NumPy, Scikit-learn)**, performing tasks like missing value **imputation**, **normalization**, **encoding**, and feature selection
- Implemented basic **ML models**, including **linear regression**, decision trees, and **K-Means clustering**, to identify trends and segment data for exploratory analysis
- Practiced data wrangling techniques to merge and transform multiple datasets from **CSV** and **Excel** sources, preparing them for analysis and modeling
- Used **Jupyter Notebooks** and **Google Collab** to document **exploratory analysis** and build reproducible code for model experimentation
- Gained exposure to the **machine learning model lifecycle**, from **data ingestion** and **cleaning** to **evaluation** using metrics like **accuracy**, **precision**, and **recall**
- Collaborated with mentors and senior data scientists on small projects to understand real-world data science **workflows**, **code versioning** with **Git**, and basic project documentation

## **GEETHANJALI COLLEGE OF ENGINEERING AND TECHNOLOGY**

**Hyderabad, India**

### **Editor in Chief**

**Dec 2019 - June 2022**

- Founded and led Technochronicle, a college magazine aimed at promoting awareness of advancements in Electronics & Communication Engineering
- Built and managed a cross-functional team to curate, edit, and publish student- and faculty-submitted articles, fostering a culture of technical writing and knowledge sharing within the department; coordinated with faculty and management for funding and approvals

## **PROJECTS**

### **Deque Systems, May 2024**

- Contributed to data analysis initiatives aimed at enhancing data integrity and establishing table relationships as part of our Capstone course, while also crafting prompts to assess the reliability of Axe Advisor, an AI tool utilized for accessibility-related topics

### **Global Maternal Healthcare Disparities' Analysis, Dec 2023**

- Implemented robust data collection strategies for a central maternal health database, ensuring accuracy and completeness
- Designed a user-centric data storage system through effective data modeling. Conducted comprehensive data analysis to enable informed decision-making and strategic planning, extracting valuable insights

### **Movies and News Data Analysis, Sept 2023**

- A machine learning project that analyzes two decades of movie data alongside contemporaneous news headlines to explore correlations between media coverage and popular film trends
- Implemented Web scraping, API data extraction, SpaCy, Naïve Bayes, KNN, K-Means clustering, Decision Trees, and Association Rule Mining

### **CDC Survey Data pipeline modeling, May 2023**

- Built a pipeline for CDC health survey data to determine patterns in eating and sleeping disorders and their association with AUD
- Implemented concepts include KNN, Logistic Regression, T-SNE feature Extraction, Gradient Boost, SVM, Risk Score predictor for AUD

### **Tableau Dashboard**

- Data analysis of 1000 years of Natural disasters sourced from EM-DAT using several statistical analysis tests and Machine

learning models

- Performed Statistical tests, Multivariate Analysis, Stationarity tests, Causality tests, Hypothesis testing, LSTM, ARIMA, and SARIMAX

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#### **EDUCATION**

**UNIVERSITY OF COLORADO**

**Master of Science in Data Science**

**BOULDER, CO**

**Aug 2022 - May 2024**

**GEETHANJALI COLLEGE OF ENGINEERING AND TECHNOLOGY**

**B.Tech in Electronics and Communication Engineering,**

**Hyderabad, India**

**July 2018 - July 2022**